Humanized LAG3 Mouse

Strain Name: B6.129-*Lag3* $^{tm1(hLAG3)/Smoc}$ Strain Background: B6.129 Cat. No. : IT-HU-00049

LAG3（lymphocyte activating 3, also known as CD223) has been shown to act as a co-inhibitory molecule expressed on activated T cells, NK cells, B cells, and plasmacytoid dendritic cells. LAG3 is an immune checkpoint receptor that binds to the antigen-MHC complex to present antigens to T cells. Experiments have shown that LAG3 negatively regulates T cell proliferation as well as the development of lasting memory T cells.

Construction strategy

The coding sequence for the extracellular domain of mouse endogenous LAG3 was completely replaced by the human LAG3 counterpart, leading to the expression of a chimeric LAG3 protein.

Validation data

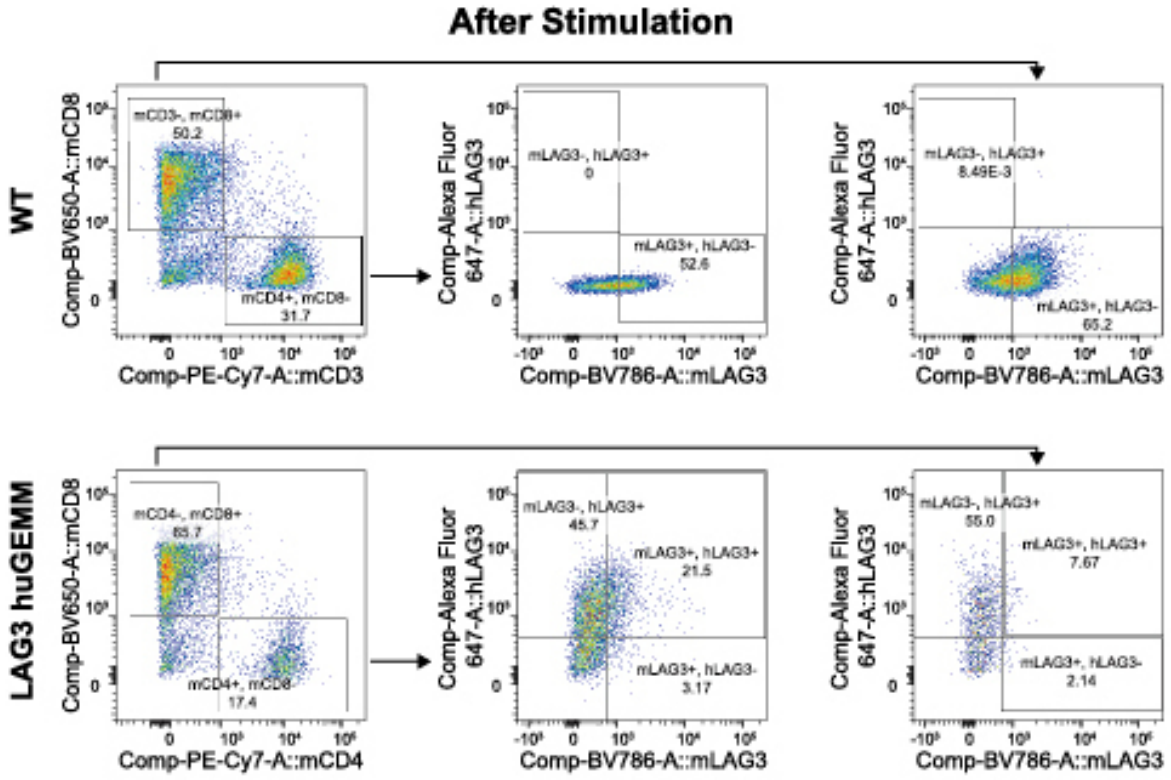


Figure 1. Expression of LAG3 in the activated spleen lymphocytes of humanized LAG3 mice is detected by FACS. Stimulated spleen lymphocytes of homozygous humanized LAG3 mice. The results showed that the active expression of humanized LAG3 can be detected in both activated CD4+ and CD8+ T lymphocytes collected from homozygous humanized LAG3 mice. (Completed in collaboration with CrownBio).

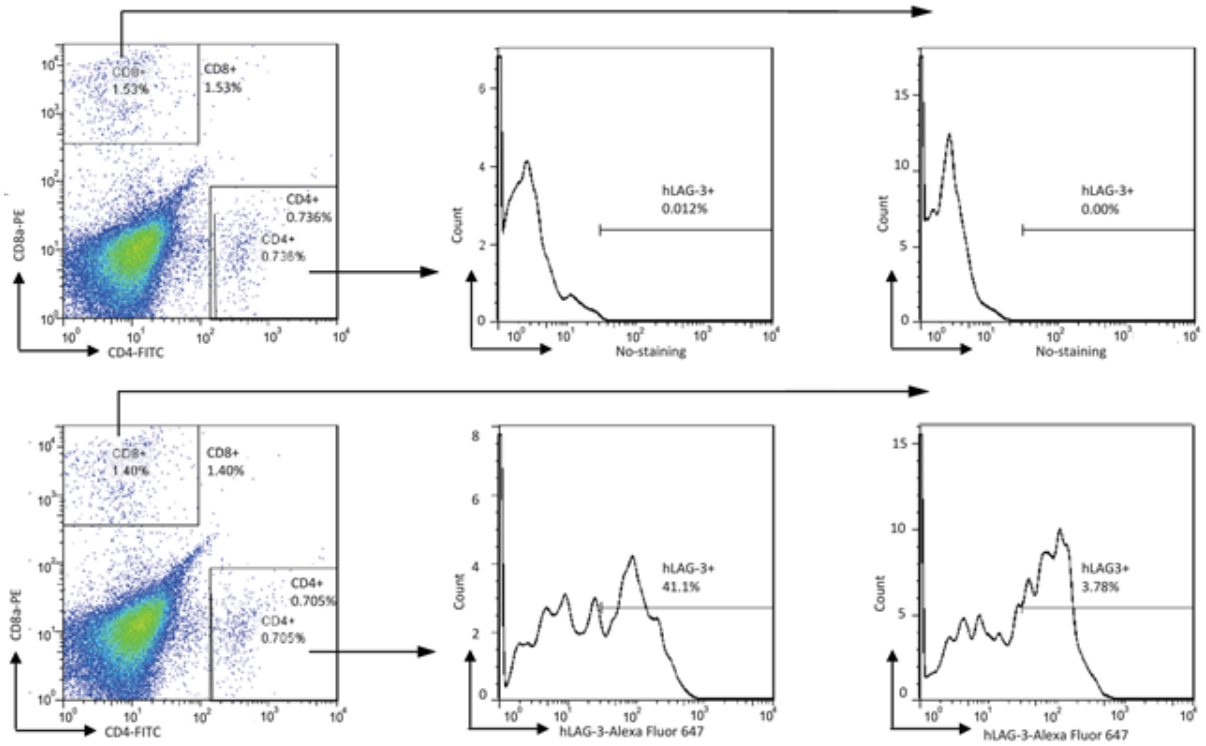
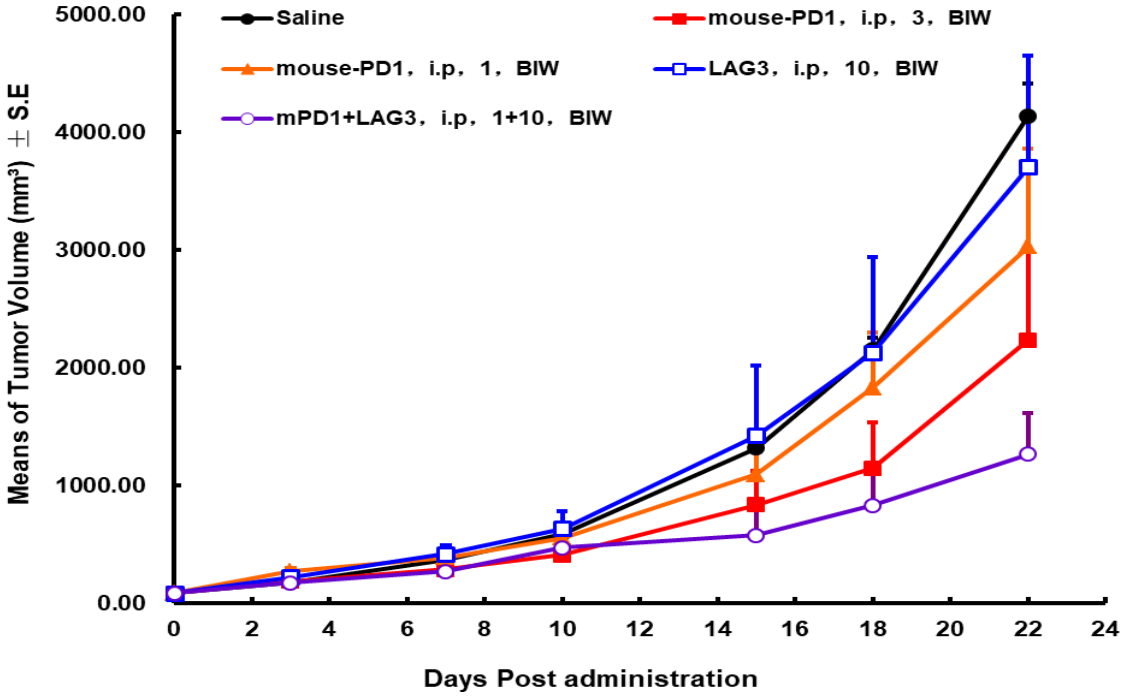


Figure 2. Flow cytometry analysis of tumor infiltrating lymphocytes (TILs) after humanized LAG3 mice were inoculated with MC38 tumor cells (data were obtained in cooperation with GenScript)

FACS detection of LAG3 expression in tumor infiltrating lymphocytes collected from humanized LAG3 mice. Homozygous humanized LAG3 mice were inoculated with MC38 colon cancer cells. After the tumors grew to 1000 mm^3, tumor infiltrating lymphocytes were isolated and detected by FACS to measure the expression of humanized LAG3 in $CD4^+$ and $CD8^+$ T cells. The results showed that the active expression of humanized LAG3 was detected in the tumor infiltrating lymphocytes collected from humanized LAG3 mice. In particular, 41.1% of $CD4^+$ positive cells were also positive for humanized LAG3.



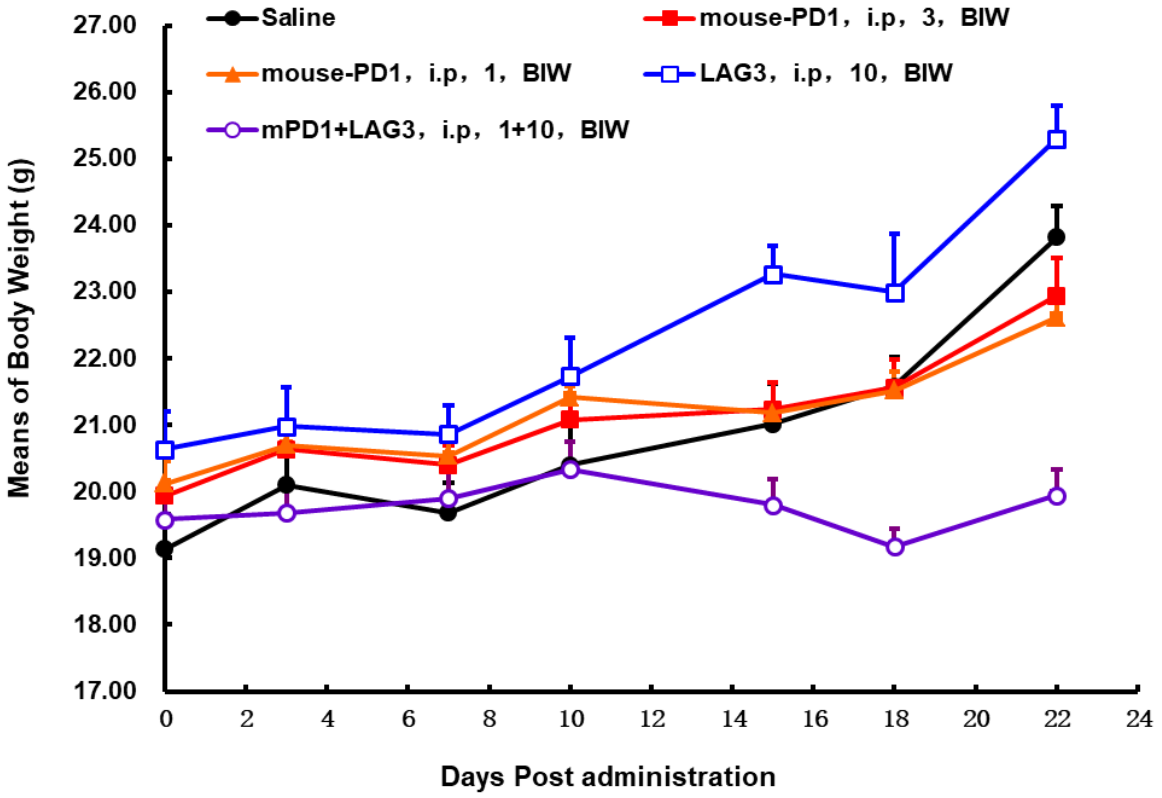
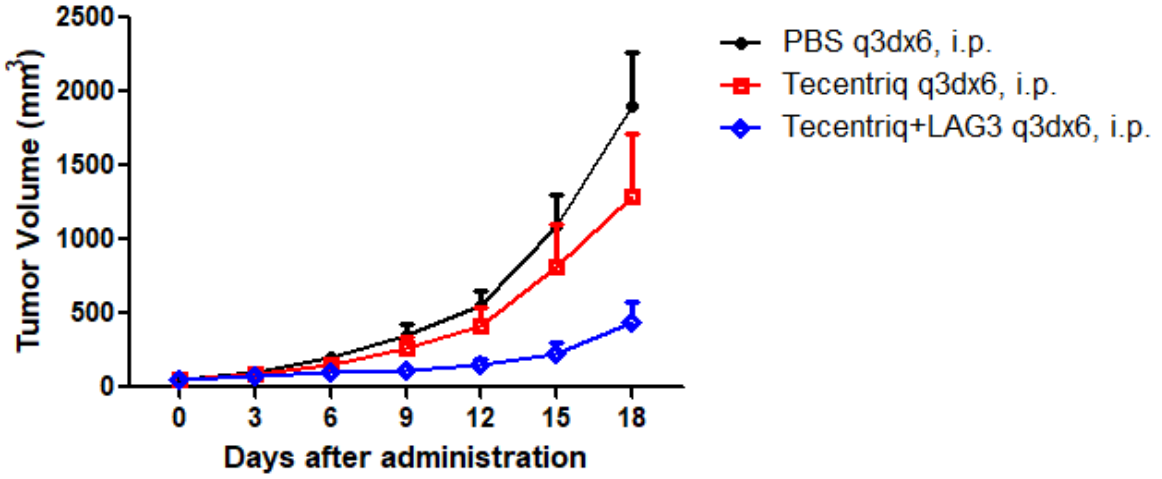


Figure 3. In vivo validation of anti-tumor efficacy in a MC38 tumor-bearing model of humanized LAG3 mouse. Homozygous humanized LAG3 mice were inoculated with MC38 colon cancer cell lines. After the tumors grew to 50 mm3, the animals were randomly assigned into different groups (n=5). The results showed that the anti-tumor effect was not observed when the antibody targeting human LAG3 was administered alone. However, a significant anti-tumor effect was observed when the antibody targeting human LAG3 was administered together with the anti-PD1 antibody.



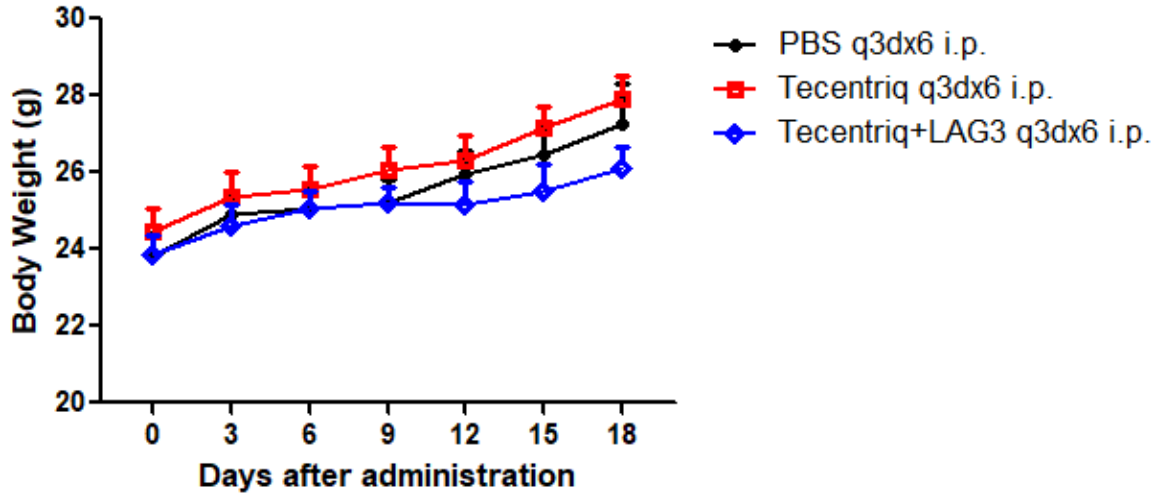


Figure 4. In vivo validation of anti-tumor efficacy in a MC38 tumorbearing model of humanized LAG3 mouse. Homozygous humanized LAG3 mice were inoculated with MC38 colon cancer cell lines. After the tumors grew to 70-80 mm3, the animals were randomly assigned into different groups (n=8). The results showed a significant anti-tumor effect was observed when the antibody targeting human LAG3 was administered together with Tencentriq.

Immune Checkpoint Humanized Mouse Models

Being recognized as a top scientific breakthrough in 2013, cancer immunotherapy is predicted to be one of the most promising research areas for improving patient outcomes. Although many immunotherapy breakthroughs may still lie ahead, important clinical advances have been made in the past few years for some of the deadliest cancers, reaffirming the potential of immunotherapy for many types of patients.

However, it is worth noting that drug candidates developed to interfere with human proteins may not comparably interact with their murine counterparts. It is therefore critical to develop humanized mouse models to enable in vivo efficacy evaluation of cancer immunotherapies.

Immune Checkpoint Humanized Mouse Models available at ingenious targeting laboratory

4-1BB
CD40
CD47
CD73 (NT5E)
CTLA4 (C57BL/6)
CTLA4 (BALB/c)
KDR
LAG3
OX40
OX40/CTLA4
PD-1 (C57BL/6)
PD-1 (BALB/c)
PD-1/4-1BB
PD-1/CD40
PD-1/CTLA4
PD-1/LAG3
PD-1/OX40
PD-1/PD-L1
PD-1/TIGIT
PD-1/TIM3
PD-L1
PD-L1/CTLA4
PD-L1/LAG3
PD-L1/OX40
PD-L1/TIGIT
SIRPA
SIRPA/CD47
TIGIT
TIM3 (C57BL/6)
TIM3 (BALB/c)
TNFRSF1B
And more to come!

To get to know more about these models, visit our website **www.genetargeting.com** or contact our scientific experts at **inquiry@genetargeting.com**

About ingenious targeting laboratory

ingenious targeting laboratory (**ingenious**) has been a leading global provider of custom genetically modified mouse, rat, and rabbit models for over 20 years. As one of the very first mouse gene targeting companies, our trusted service is built on two decades' worth of successful animal model creation for investigators, organizations, and companies worldwide. Our models have been published in hundreds of journals including *Science*, *Nature*, and *Cell*, making us one of the most validated and respected production companies in the industry. We are excited to add catalog mouse models to our service repertoire by means of our collaboration with Shanghai Model Organisms Center (SMOC).

ingenious targeting laboratory

2200 Smithtown Ave, Ronkonkoma, NY 11779

www.genetargeting.com • inquiry@genetargeting.com